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AMPLIFIER WITH COMMON MODE GAIN REDUCTION

Abstract

An amplifier circuit includes a first transistor having a terminal and a second transistor having a terminal coupled to the terminal of the first transistor. A third transistor has a first terminal coupled to the terminals of the first and second transistors and has a control terminal. A fourth transistor has a control terminal coupled to the terminals of the first and second transistors and to the first terminal of the third transistor. The fourth transistor has a second terminal. A fifth transistor has a control terminal coupled to the second terminal of the fourth transistor and has second and third terminals. A sixth transistor has a control terminal coupled to the control terminal of the third transistor and to the third terminal of the fifth transistor. The sixth transistor has a second terminal coupled to the second terminal of the fifth transistor.

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Background/Summary

BACKGROUND

[0001] Numerous systems employ differential signaling which includes a pair of identical, but opposite phase, signals. Many systems may include a differential amplifier to amplify a differential input voltage. The common mode gain of an amplifier is the ratio of the change in the common mode voltage at the amplifier's output to a change in the common mode voltage at the amplifier's input. The design of differential amplifiers may take into account multiple factors including, for example, the power consumption of the amplifier, the noise produced by the amplifier, and the common mode gain of the amplifier.

SUMMARY

[0002] In one example, an amplifier circuit includes a first transistor having a terminal and a second transistor having a terminal coupled to the terminal of the first transistor. A third transistor has a first terminal coupled to the terminals of the first and second transistors and has a control terminal. A fourth transistor has a control terminal coupled to the terminals of the first and second transistors and to the first terminal of the third transistor. The fourth transistor has a second terminal. A fifth transistor has a control terminal coupled to the second terminal of the fourth transistor and has second and third terminals. A sixth transistor has a control terminal coupled to the control terminal of the third transistor and to the third terminal of the fifth transistor. The sixth transistor has a second terminal coupled to the second terminal of the fifth transistor.

[0003] In another example, a data converter includes a first stage circuit, a second stage circuit, and an amplifier. The first stage circuit has an output. The second stage circuit has an input coupled to the output of the first stage circuit. The amplifier has an input coupled to the output of the first stage and has an output coupled to the input of the second stage. The amplifier includes first, second, third, fourth, fifth, and sixth transistors. The first transistor has a first terminal and a control terminal. The control terminal of the first transistor is coupled to the input of the amplifier. The second transistor has a first terminal coupled to the first terminal of the first transistor and has a control terminal coupled to the input of the amplifier. The third transistor has a first terminal coupled to the first terminals of the first and second transistors and has a control terminal. The fourth transistor has a control terminal coupled to the first terminals of the first and second transistors and to the first terminal of the third transistor. The fourth transistor has a second terminal. The fifth transistor has a control terminal coupled to the second terminal of the fourth transistor and has second and third terminals. The sixth transistor has a control terminal coupled to the control terminal of the third transistor and to the third terminal of the fifth transistor. The sixth transistor has a second terminal coupled to the second terminal of the fifth transistor.

[0004] In yet another example, a data converter includes a first stage circuit, a second stage circuit, and an amplifier. The first stage circuit has an output. The second stage circuit has an input coupled to the output of the first stage circuit. The amplifier has an input coupled to the output of the first stage circuit and has an output coupled to the input of the second stage. The amplifier includes first, second, and third transistors and a tail current control circuit. The first transistor has a first terminal and has a control terminal coupled to the input of the amplifier. The second transistor has a first terminal coupled to the first terminal of the first transistor and has a control terminal coupled to the input of the amplifier. The third transistor has a first terminal coupled to the first terminals of the first and second transistors and has a control terminal. The tail current control circuit has a first terminal coupled to the control terminal of the third transistor and has a second terminal coupled to the first terminals of the first and second transistors. The tail current control circuit is configured to adjust a tail current through the third transistor based on a voltage at the first terminals of the first and second transistors.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a block diagram of a data converter, in an example.

[0006] FIG. 2 is a schematic diagram of an amplifier usable in the data converter of FIG. 1, the amplifier including tail current control circuits, in an example.

[0007] FIG. 3 is a schematic diagram of the tail current control circuit of FIG. 2, in an example.

[0008] FIG. 4 is a schematic diagram of the tail current control circuit of FIG. 2, in another example.

[0009] FIG. 5 is a schematic diagram of the tail current control circuit **250** useful at higher frequencies, in an example.

[0010] FIG. 6 is a schematic diagram of a tail current control circuit including the features of the examples of FIGS. 3, 4, and 5 to reduce the common gain across a wide range of frequencies.

[0011] FIG. 7 is a schematic diagram of tail current control circuit, in another example.

DETAILED DESCRIPTION

[0012] The same reference numbers or other reference designators are used in the drawings to designate the same or similar (either by function and/or structure) features.

[0013] The examples described herein pertain to a dynamic differential amplifier with a relatively low common mode gain. Such amplifiers may be used, for example, in a data converter, such as an analog-to-digital converter (ADC). However, the principles described herein may apply to other types of differential amplifiers as well as other types of systems besides data converters.

[0014] FIG. 1 is a block diagram of an ADC **100**, in an example. In this example, ADC **100** is a pipeline converter including stages **110**, **120**, and **130**, amplifiers A2 **162** and A3 **172**, a digital error correction circuit **140**, a reference voltage circuit **142**, and a timing circuit **144**. ADC **100** includes differential input terminals **101** and has a digital output terminal **195**. ADC **100** converts a differential analog input voltage (AIN-AIN) provided at its differential input terminals **101** into a multi-bit (e.g., 13 bits) digital output value, D at its digital output terminal **195**. Stage **110** includes an amplifier A1 **111**, a track and hold circuit (TH1) **112**, a flash analog-to-digital converter ADC1 **115**, a digital-to-analog converter (DAC) DAC1 **116**, and a summer **114**. Amplifier A1 has a differential input **111a** coupled to differential input terminals **101** and has a differential output **111b** coupled to the input of TH1 **112**. TH1 **112** converts a continuous time signal at its input to a sampled domain signal at its output. The output of TH1 is coupled to a positive (+) input of summer **114** and to an input of flash ADC1 **115**. Flash ADC1 **115** has an output coupled to an input of DAC1 **116**, and the output of DAC1 **116** is coupled to a negative (−) input of summer **114**. Flash ADC1 **115** may be a low-resolution ADC, e.g., a one, two, or three-bit ADC, and, accordingly, provides a relatively low-resolution digital equivalent (bits **113**) of the output of amplifier A1 **111** during a sample phase of stage **110**. The digital DAC1 **116** converts the digital value represented by bits **113** from flash ADC1 **115** to an analog equivalent signal of the digital equivalent. Summer **114** subtracts the analog signal from DAC1 **116** from the analog output of TH1 **112** to produce a residue voltage RES1 at its output. Bits **113** from ADC1 **115** are provided to digital error correction circuit **140**.

[0015] The residue signal RES1 is provided to an input **162a** (e.g., a differential input) of amplifier A2 **162**. During a hold phase, amplifier **162** amplifies the residue signal RES1 from stage **110**. The output **162b** of amplifier A2 is coupled to stage **120**. Stage **120** is configured similar to stage **110**. Stage **120** includes a flash ADC2 **125**, a DAC2 **126**, and a summer **124**. The amplified residue signal from amplifier **162** is provided to the positive input of summer **124** and to an input of flash ADC2 **125**. During a sample phase for stage **120**, flash ADC2 **125** produces additional bits **123** (e.g., 1, 2, 3, etc.) bits of the digital output code and provides those bits to digital error correction circuit **140**. The output of flash ADC2 **125** is also coupled to an input of DAC2 **126**, and the output of DAC2 **126** is coupled to the negative input of summer **124**. DAC2 **126** converts the digital value represented by bits **123** to an analog signal equivalent. Summer **124** produces another residue

signal RES2 based on the differences between the amplified residue from the first stage **110** and the analog signal from DAC2 **126**.

[0016] Amplifier A3 **172** includes an input **172a** (e.g., a differential input) and an output **172b**. Input **172a** is coupled to the output of summer **124**. During a hold phase, amplifier A3 **172** amplifies the residue signal RES2 from second stage **120** and provides an amplified residue signal to third stage **130**. In the example of FIG. **1**, third stage **130** includes ADC3 **130**, which converts the amplified residue from amplifier A3 to the remaining (e.g., lowest order) bits **133** of the digital code equivalent of the differential input signal AIN/AIN. In some examples, ADC3 **130** may include a delay domain ADC.

[0017] Digital error correction circuit **140** provides one or more corrections to the digital code represented by bits **113**, **123**, and **133** to improve (e.g., increase) the signal-to-noise ratio (SNR) of ADC **100**. Digital error correction circuit **140** provides the corrected digital code to digital output terminal **195**.

[0018] Reference voltage circuit **142** produces one or more reference voltages for use by ADC1 **115**, ADC2 **125**, and ADC3 in stage **130**. Timing circuit **144** produces one or more clock signals (clocks) for use to control the various circuits of ADC **100**. For example, timing circuit **144** generates clocks CLK_PCAS, CLK_NCAS, CLK_AMPON, CLK1, and CLK1, whose uses are shown and described below with respect to FIG. **2**. Clock CLK1 is an inverse of CLK1 (e.g., 180 degrees out of phase with respect to each other). Timing circuit **144** may generate a separate clock CLK_AMPON for each of amplifiers A2 and A3.

[0019] In some examples, amplifier A1 is a continuous-time amplifier, and amplifiers A2 and A3 are dynamic integrating amplifiers. A continuous-time amplifier continuously amplifies its input signal. A dynamic integrating amplifier is turned on during each hold phase and turned off when not in the hold phase.

[0020] FIG. **2** is a schematic diagram of an amplifier **200**, in an example. Amplifier **200** can be used to implement, for example, either or both of amplifiers A2 **162** or A3 **172**, which, as noted above, are dynamic integrating amplifiers. Amplifier **200** includes an input stage **208**, switches SW1a and SW1b, a switched capacitor common-mode feedback (CMFB) circuit **220**, tail current control circuits **250** and **252**, and transistors M3, MN3, MPCAS1, MPCAS2, MNCAS1, MNCAS2, MCS1, and MCS2. Input stage **208** includes transistors M1, M2, MN1, and MN2. In this example, transistors M1, M2, M3, MPCAS1, and MPCAS2 are p-channel field effect transistors (PFETs) and transistors MN1, MN2, MN3, MNCAS1, MNCAS2, MCS1, and MCS2 are n-channel field effect transistors (NFETs). Capacitance Cpara between the gates of transistors MCS1/MCS2 and the voltage supply terminal **209** represents a parasitic capacitance of transistors MCS1 and MCS2 including, for example, their gate-to-source capacitance and drain-to-source capacitance.

[0021] The source terminals (or sources) of transistors M1 and M2 are coupled together and, through switch SW1a, to the drain terminal (or drain) of transistor M3. Similarly, the sources of transistors MN1 and MN2 are coupled together and, through switch SW1b, to the drain of transistor MN3. Switches SW1a and SW1b are controlled based on clock CLK_AMPON from timing circuit **144**. Amplifier **200** in this example is a dynamic integrating amplifier, which is turned on during each hold phase and turned off when not in the hold phase. When switches SW1a and SW1b are closed (turned on), amplifier **200** is on and amplifies its differential input, INP-INM, to produce a differential output, OUTP-OUTM. When switches SW1a and SW1b are open (turned off), the amplifier is off and consumes little or no power. Accordingly, amplifier **200** can be turned off to save power. In one example, each of amplifiers A2 **162** and A3 **172** can be turned off during a portion of each cycle of the respective stage **110** and **120**. For example, amplifier A2 **162** is turned off while stage **110** is beginning to generate residue signal RES1. Upon residue signal RES1 settling, timing circuit **144** asserts clock signal CLK_AMPON to a logic state to turn on amplifier A2 **162** to cause amplifier A2 to amplify RES1. Similarly, timing circuit **144** turns on amplifier A3 upon residue signal RES2 from stage **120** settling.

[0022] The current through transistor M3 to the sources of transistors M1 and M2 is tail current $I_{sub.ptail}$. Current $I_{sub.ptail}$ is the sum of a direct current (DC) current plus a “small signal” current $i_{sub.ptail}$. Small signal current $i_{sub.ptail}$ represents the change in the tail current $I_{sub.ptail}$ due to a change in the common mode voltage. Similarly, current $I_{sub.ntail}$ through transistor MN3 also includes a DC component and a small signal component, $i_{sub.ntail}$. While DC tail currents should persist through transistors M3 and MN3 when switches SW1a and SW1b are closed, the small signal current components, $i_{sub.ptail}$, and $i_{sub.ntail}$ should be small or zero. A small or zero common mode gain for the amplifier means that currents $i_{sub.ptail}$ and $i_{sub.ntail}$ advantageously are small or zero.

[0023] The sources of transistors M3 and MN3 are coupled to respective voltage supply terminals 207 (e.g., +Vdd) and 209 (e.g., -Vdd). The differential input to amplifier 200 includes terminals 201 and 202. Terminals 201 and 202 may correspond to input 162a of amplifier A2 162 and/or input 172a of amplifier A3 172. The gate terminals (or gates) of transistors M1 and MN1 are coupled together and to amplifier terminal 201. The gates of transistors M2 and MN2 are coupled together and to amplifier terminal 202. The drains of transistors M1 and MN1 are coupled together and are one output 221 of the input stage 208. Similarly, the drains of transistors M2 and MN2 are coupled together and are another output 223 of input stage 208. Output 221 is coupled to the source of transistor MPCAS1, and output 223 is coupled to the source of transistor MPCAS2.

[0024] The drains of transistors MPCAS1 and MNCAS1 are coupled together at amplifier output 211. The signal at amplifier output 211 is OUTP. An output capacitor C0_1 is coupled between amplifier output 211 and voltage supply terminal 209. Similarly, the drains of transistors MPCAS2 and MNCAS2 are coupled together at amplifier output 212. The signal at amplifier output 212 is OUTN. An output capacitor C0_2 is coupled between amplifier output 212 and voltage supply terminal 209. Amplifier outputs 221 and 212 may correspond to output 162b of amplifier A2 162 and/or out 172b of amplifier A3 172. The capacitance of capacitors C0_1 and C0_2 may be the same and is referred to herein as “C0.” The source of transistor MNCAS1 is coupled to the drain of transistor MCS1, and the source of transistor MNCAS2 is coupled to the drain of transistor MCS2. The sources of transistors MCS1 and MCS2 are coupled to voltage supply terminal 209.

[0025] The gates of transistors MPCAS1 and MPCAS2 are coupled together and receive clock CLK_PCAS from the timing circuit 144. The gates of transistors MNCAS1 and MNCAS2 are coupled together and receive clock CLK_NCAS from the timing circuit 144. The gates of transistors MCS1 and MCS2 are coupled together and to switched capacitor CMFB circuit 220. In the example of FIG. 2, switched capacitor CMFB circuit 220 includes a capacitor network which includes capacitors C_221, C_222, C_223, and C_224 and switches SW_221, SW_222, SW_223, SW_224, SW_225, and SW_226. Switches SW_221 and SW_224 are coupled in series between amplifier output 212 and a reference terminal 231 having a reference voltage OUTCM_REF. Similarly, switches SW_223 and SW_226 are coupled in series between amplifier output 211 and a reference terminal 232 also having reference voltage OUTCM_REF. Switches SW_222 and SW_225 are coupled in series between the gates of transistors MCS1 and MCS2 and terminal 233 which has another reference voltage BIAS_REF. Capacitor C_221 is coupled between amplifier output 211 and the gates of transistor MCS1 and MCS2. Capacitor C_222 is coupled between the gates of transistor MCS1 and MCS2 and amplifier output 212. Capacitors C_223 and C_224 are coupled in series as shown. Switches SW_221, SW_222, and SW_223 are operated according to clock CLK1, and switches SW_224, SW_225, and SW_226 are operated according to clock CLK1. Switched capacitor CMFB circuit 220 samples the common mode voltage at the output of amplifier 200, compares the sensed common mode output voltage to a reference voltage, and provides a voltage to the gates of transistors MCS1 and MCS2 to reduce or increase the output common-mode current at the output of amplifier 200 to thereby maintain a steady output common mode voltage.

[0026] Each transistor in FIG. 2 can be characterized according to one or more transconductance parameters. For example, transconductance $g_{sub.m}$ for a transistor is the ratio of a change in its

drain current (Id) due to a change in the gate-to-source voltage (Vgs). Transconductance g.sub.ds for a transistor is the ratio of its drain current due to a change in its drain-to-source voltage (Vds). FIG. 2 lists various transconductances for its transistors. For example, the g.sub.m and g.sub.ds for transistor M3 is g.sub.m_ptail and g.sub.ds_ptail, respectively. The g.sub.m and g.sub.ds for transistor MN3 is g.sub.m_ntail and g.sub.ds_ntail, respectively. The g.sub.m for transistors M1 and M2 is g.sub.mp (the same g.sub.m value). The g.sub.m for transistors MN1 and MN2 is g.sub.mn. The g.sub.m for transistors MPCAS1 and MPCAS2 is g.sub.m_pcas, and the g.sub.m for transistors MNCAS1 and MNCAS2 is g.sub.m_ncas. The g.sub.m for transistors MCS1 and MCS2 is g.sub.mcs.

[0027] ADC3 130 may be a delay domain ADC as noted above. Delay domain ADCs have relatively low common mode rejection ratios (CMRRs) which makes them sensitive to variation in input common mode voltage. Amplifier 200, if used to implement, for example, amplifiers A2 162 and/or A3 172, advantageously generates relatively little noise and consumes relatively little power. However, as explained below, absent the tail current control circuits 250 and 252, amplifier 200 may not have a sufficiently low common mode gain (A.sub.cm) to balance out the relatively low CMRR of delay domain ADC3 130.

[0028] The common mode gain of amplifier 200 can be derived based on the following equations.

$$[00001] G_{m_out} = \left(\frac{g_{mp}}{1 + g_{mp}/g_{ds_ptail}} \right) + \left(\frac{g_{mn}}{1 + g_{mn}/g_{ds_ntail}} \right) \quad (\text{Eq. 1})$$

$$G_{in} = \frac{g_{dsp} * g_{ds_ptail}}{g_{mp}} + \frac{g_{dsn} * g_{ds_ntail}}{g_{mn}} \quad (\text{Eq. 2}) \quad G_{outp} = \left(\frac{g_{ds_pcas}}{g_{m_pcas}} \right) G_{in} \quad (\text{Eq. 3}) \quad G_{outn} = g_{mcs} * A \quad (\text{Eq. 4})$$

where

$$[00002] A = \left(\frac{2C}{2C + C_{para}} \right)$$

and C is the capacitance of capacitors C_221 through C_224 of the switched capacitor CMFB circuit 220.

$$[00003] G_{out} = G_{outp} + G_{outn} \quad (\text{Eq. 5}) \quad A_{cm} = \frac{G_{m_out}}{C_o} (1 - e^{-T_{int}/\tau}) \quad (\text{Eq. 6})$$

where, T.sub.int is the integration time of amplifier 200 and $\tau = C_{sub.o}/G_{sub.out}$.

[0029] The open loop gain (A.sub.oL) of the switched capacitor CMFB circuit 220 is given by:

$$[00004] A_{oL} = \left(\frac{g_{mcs} * T_{int}}{C_o} \right) A \quad (\text{Eq. 7})$$

The value of A.sub.oL may be relatively low. For example, for illustrative values of g.sub.mcs of 24 mS, a T.sub.int of 140 ps, C.sub.o of 2.9 pF, C of 200 fF, and C.sub.para of 180 fF, A.sub.oL will be equal to 1.2 dB which is very low. The common mode gain A.sub.cm is given by:

$$[00005] A_{cm} = G_{m_out} * R_0 [1 - \exp(-\frac{T_{int}}{R_0} * C_o)] \quad (\text{Eq. 9})$$

The output resistance R.sub.o of amplifier is the output resistance without the switched capacitor CMFB circuit 220 divided by (1+A.sub.oL),

$$[00006] R_0 = \frac{R_o}{1 + A_{oL}} \quad (\text{Eq. 10})$$

For integration times that are less than the product of R.sub.o and C.sub.o

(T.sub.int << R.sub.o * C.sub.o),

$$[00007] A_{cm} \approx G_{m_out} * T_{int} / C_o \quad (\text{Eq. 11})$$

For integration times that are more than the product of R.sub.o and C.sub.o

(T.sub.int >> R.sub.o * C.sub.o), then

$$[00008] A_{cm} \approx G_{m_out} * R_0 \quad (\text{Eq. 12})$$

Accordingly, if R.sub.o is high enough such that T.sub.int << R.sub.o * C.sub.o, then per Eq. (11), A.sub.cm can be reduced by reducing G.sub.m_out because T.sub.int and C.sub.o are fixed by other requirements such as gain, power consumption, and noise. But, if R.sub.o is relatively low such that T.sub.int >> R.sub.o * C.sub.o, then per Eq. (12), A.sub.cm will be proportional to R.sub.o

and, accordingly, as $R_{sub.0}$ is reduced, $A_{sub.cm}$ also reduces. The switched capacitor CMFB circuit **220** is used to reduce the output resistance $R_{sub.0}$ per Eq. (10). However, the switch capacitor CMFB circuit **220** may not be able to reduce the output resistance $R_{sub.0}$ enough to get a sufficiently low $A_{sub.cm}$, and other techniques to reduce $A_{sub.cm}$ may be employed, as described below.

[0030] Tail current control circuits **250** and **252** are coupled to the respective gates of transistors **M3** and **MN3**. Terminal **250a** of tail current control circuit **250** is coupled to the gate of transistor **M3**, and terminal **252a** of tail current control circuit **252** is coupled to the gate of transistor **MN3**. Terminal **250b** of tail current control circuit **250** is coupled to the sources of transistors **M1** and **M2**, and terminal **252b** of tail current control circuit **252** is coupled to the sources of transistors **MN1** and **MN2**. As described below, tail current control circuits **250** and **252** cause amplifier **200** to have a lower common mode gain than it otherwise would have without the tail current control circuits.

[0031] FIG. 3 is a schematic diagram of an example tail current control circuit **250** usable for amplifiers **A2 162** and/or **A3 172**. In this example, tail current control circuit **250** includes transistors **M4**, **M5**, and **M6**, a sampling circuit **314**, and current source circuits **IBIAS1** and **IBIAS2**. Capacitor C_{decap} is a capacitor between the source and gate of transistor **M3** and may be included to reduce the noise of amplifier **200** which otherwise may occur due to fluctuations or switching noise on the power supply voltage at the voltage supply terminals **207**. Capacitance $C_{sub.dg_ptail}$ is the parasitic drain-to-source capacitance of transistor **M3**. Sampling circuit **314** includes a capacitor **C3** and a switch **SW2**. Transistor **M4** is an NFET and transistors **M5** and **M6** are PFETs. The gate of transistor **M6** is coupled to the gate of transistor **M3** and to the drain of transistor **M5**. The source of transistor **M6** is coupled to the voltage supply terminal **207**. The drain of transistor **M6** is coupled to the source of transistor **M5**. The $g_{sub.m}$ and $g_{sub.ds}$ of transistor **M6** is designated as $g_{sub.m,bias}$ and $g_{sub.ds,bias}$, respectively, as shown in FIG. 3. The voltage on the gates of transistors **M3** and **M6** is $V_{sub.g_ptail}$. A terminal of current source circuit **IBIAS2** is coupled to the drain of transistor **M5**, and another terminal of current source circuit **IBIAS2** is coupled to the voltage supply terminal **209**. The gate of transistor **M5** is coupled to the source of transistor **M4**. A terminal of current source circuit **IBIAS1** is coupled to the source of transistor **M4**, and another terminal of current source circuit **IBIAS1** is coupled to the voltage supply terminal **209**. The drain of transistor **M4** is coupled to the voltage supply terminal **207**. Switch **SW2** is coupled between the sources of transistors **M1/M2** and the gate of transistor **M4**. Capacitor **C3** is coupled between the gate of transistor **M4** and the voltage supply terminal **209**.

[0032] The voltage at the gate of transistors **M1** and **M2** is the input common mode voltage $V_{sub.cm}$ plus or minus the differential input voltage $V_{sub.d}$. The voltage at the gate of transistor **M1** is $V_{sub.cm} + V_{sub.d}$, and the voltage at the gate of transistor **M2** is $V_{sub.cm} - V_{sub.d}$. Per a small signal analysis, transistors **M1** and **M2** function as source followers. Accordingly, the voltage at the sources of transistors **M1** and **M2** is proportional to the common mode voltage. The voltage at the sources of transistors **M1** and **M2** is sampled by sampling circuit **314**, e.g., by charging capacitor **C3** to the voltage of the sources of transistors **M1** and **M2** when switch **SW2** closes. Switch **SW2** is controlled based on the same signal **CLK_AMPON** that closes switch **SW1a**.

[0033] The combination of transistor **M4** and current source circuit **IBIAS1** is such that transistor **M4** is configured as a source-follower circuit. Accordingly, the voltage $v_{sub.ts}$ at the source of transistor **M4** is the gate voltage of transistor **M4** minus the threshold voltage $V_{sub.th}$ of transistor **M4**. Consequently, voltage $V_{sub.ts}$ is proportional to the common mode voltage $V_{sub.cm}$. Voltage $v_{sub.ts}$ also is the voltage at the gate of transistor **M5**.

[0034] The tail current control circuit **250** in FIG. 3 is explained with respect to a small signal analysis. Accordingly, the small signal tail current $i_{sub.ptail}$ is labeled in FIG. 3. A change in the common mode voltage $V_{sub.cm}$ causes a change in current $i_{sub.ptail}$. For example, an increase in $v_{sub.cm}$ causes the small signal voltage $v_{sub.s,amp}$ on the sources of transistors **M1** and **M2** to increase. With respect to the $g_{sub.ds_ptail}$ of transistor **M3**, tail current $i_{sub.ptail}$ can be

determined as:

$$[00009] \ i_{ptail} = v_{s,amp} * g_{ds_ptail} \quad (Eq. 13)$$

Small signal voltage $v_{sub,s,amp}$ is sampled by sampling circuit **314**, e.g., by charging capacitor C3 to voltage $v_{sub,s,amp}$ when switch SW2 closes during the hold phase of the ADC. Voltage $v_{sub,ts}$ is the sampled voltage $v_{sub,s,amp}$ less the threshold voltage of transistor M4. Voltage $v_{sub,ts}$ is provided to the gate to transistor M5. As voltage $v_{sub,ts}$ increases, which will occur if $v_{sub,cm}$ increases, the voltage $v_{s,bias}$ (voltage on the source of transistor M5) also increases. The drain current through transistor M6 is proportional to the product of $g_{sub,ds,bias}$ and the voltage $v_{sub,s,bias}$ on the drain of transistor M6. Accordingly, an increase of the voltage on the drain of transistor M6 would tend to cause the drain current of transistor M6 to increase. However, the fixed current from current source circuit IBIAS2 prevents current through transistor M6 from changing. Accordingly, any change in the current through transistor M6 due to an increase in $v_{s,bias}$ will be balanced by an opposite polarity change in the current through transistor M6 resulting from its $g_{sub,m,bias}$ transconductance. If the current through transistor M6 attempts to increase as a result of an increase in voltage $v_{sub,s,bias}$, the voltage v_{sub,g_ptail} on the gate of transistor M6 will decrease to a level resulting in approximately no change in the small signal drain current through transistor M6. The voltage v_{sub,g_ptail} on the gate of transistor M6 is also the gate voltage for transistor M3. With respect to the g_{sub,m_ptail} transconductance of transistor M3, current $i_{sub,ptail}$ is proportional to the product of g_{sub,m_ptail} and v_{sub,g_ptail} . A decrease in voltage v_{sub,g_ptail} causes a decrease in the current $i_{sub,ptail}$ through transistor M3. Accordingly, an increase in $v_{sub,cm}$ causes an increase in $i_{sub,ptail}$ and an increase in voltage $v_{sub,ts}$. The increase in voltage $v_{sub,ts}$ causes a decrease in voltage v_{sub,g_ptail} which then reduces the drain current through transistor M3 based on its g_{sub,m_ptail} thereby advantageously counteracting the increase in current $i_{sub,ptail}$ resulting from the increase in $v_{sub,cm}$. Similarly, a decrease in the common mode voltage $V_{sub,cm}$ causes a decrease in current $i_{sub,ptail}$, which through tail current control circuit **250** causes an increase in voltage v_{sub,g_ptail} to thereby counteract the decrease in current $i_{sub,ptail}$ and maintain $i_{sub,ptail}$ relatively constant

[0035] The voltage that is sampled by sampling circuit **314** to control the loop formed by tail current control circuit **250** is the voltage at the sources of transistors M1 and M2 during the ADC's hold phase. That voltage is the residue voltage, which advantageously is a relatively small voltage. The large differential signal across inputs INP and INM is not sensed or otherwise provided to tail current control circuit **250**. The configuration of tail current control circuit **252** is the same or similar as tail current control circuit **250** and is shown and described below with respect to FIG. 7.

[0036] Because amplifier **200** is a dynamic amplifier (switch SW1a, SW1b turns on an off per CLK_AMPON), the ratio of $C_{sub,dg,sub,ptail}/(C_{sub,decap}+C_{sub,dg_ptail})$ results in some current $i_{sub,ptail}$ remaining despite the operation of the tail current control circuit **250** in FIG. 3. The gain of tail current control circuit **250** in FIG. 3 may not be sufficiently large to further reduce the current $i_{sub,ptail}$. The gain of the loop formed by transistors M4-M6 in tail current control circuit **250** in FIG. 3 is given by the ratio $g_{sub,ds,bias}/g_{sub,m,bias}$. The remaining current $i_{sub,ptail}$ can be further reduced through the example tail current control circuit **250** of FIG. 4.

[0037] FIG. 4 is a schematic diagram of tail current control circuit **250** in another example. In the example of FIG. 4, tail current control circuit **250** includes transistors M4, M5, and M6, current source circuits IBIAS1 and IBIAS, and sampling circuit **314**, as described above. The $g_{sub,m}$ and $g_{sub,ds}$ of transistor M6 is relabeled in FIG. 4 as $g_{sub,m,bias1}$ and $g_{sub,ds,bias1}$, respectively. Tail current control circuit **250** also includes transistors M7, M8, M9, and current source circuit IBIAS3. In this example, transistors M7-M9 are PFETs. The $g_{sub,ds}$ of transistor M7 is $g_{sub,ds,bias2}$. The sources of transistors M6-M8 are coupled together. The drains of transistors M6 and M7 are coupled together and to the source of transistor M5. The gates of transistors M7 and M8 are coupled together and to the drain of transistor M9. The drain of transistor M8 is coupled to

the source of transistor **M9**. A terminal of current source circuit **IBIAS3** is coupled to the drain of transistor **M9** and another terminal of current source circuit **IBIAS3** is coupled to the voltage supply terminal **209**.

[0038] A fixed bias voltage **INCM** is provided to the gate of transistor **M9**. The combination of transistors **M8** and **M9** and current source circuit **IBIAS3** generates a fixed bias voltage on the gate of transistor **M7**. The gain of the loop formed by transistors **M4-M6** in tail current control circuit **250** for the example of FIG. **4** is given by:

$$[00010] \text{ Gain of loop } = \frac{g_{ds, \text{bias}}}{g_{m, \text{bias1}}} \quad (\text{Eq. 14})$$

The value of $g_{sub.ds, \text{bias}}$ in Eq. (14) is the sum of $g_{sub.ds, \text{bias2}}$ of transistor **M7** and $g_{sub.ds, \text{bias1}}$ of transistor **M6**. The gain of the loop formed by transistors **M4-M6** in FIG. **4** is advantageously larger than the gain of the corresponding loop in FIG. **3** due to the inclusion of transistor **M7** coupled to transistor **M6** thereby causing a further reduction in current $i_{sub.ptail}$.

[0039] The tail current control circuits **250** in the examples of FIGS. **3** and **4** work well at lower frequencies but not work as well at higher frequencies due to the presence of the relatively large capacitance of capacitor **C.sub.decap**. FIG. **5** is a schematic diagram of a tail current control circuit **250** which may work better at higher frequencies than the tail current control circuits of FIGS. **3** and **4**. Tail current control circuit **250** in FIG. **5** includes transistors **M4**, current source circuit **IBIAS1** and sampling circuit **314**, as described above with respect to FIG. **3**. Tail current control circuit **250** in FIG. **5** also includes a fixed voltage circuit **510**, transistors **M10**, **M11**, **M12**, and **M15**, current source circuit **IBIAS4**, resistors **R1** and **R2**, capacitors **C4**, **C5**, and **C6**, and current source **IBIAS4**. Transistors **M11** and **M15** are PFETs and transistor **M12** is an NFET.

[0040] Fixed voltage circuit **510** includes a current source circuit **IBIAS5** and transistors **M13** and **M14** (both NFETs) coupled in series between voltage supply terminals **207** and **209**. The voltage on the source of transistor **M13** and the drain of transistor **M14** is a fixed voltage provided to the source of transistor **M12**. Transistor **M11** is a diode-connected transistor with its source coupled to the voltage supply terminal **207** and its drain and gate coupled to the drain of transistor **M12**. The source of transistor **M4** is coupled to the gate of transistor **M12**, which thus receives voltage $v_{sub.ts}$, described above. Current source circuit **IBIAS4** is coupled to the source of transistor **M10** and provides a bias current for transistor **M10**. One terminal of capacitor **C4** is coupled to the source of transistor **M10**, and the other terminal of capacitor **C4** is coupled to the gate of transistor **M15**. A terminal of resistor **R1** is coupled to the gate of transistor **M15**, and another terminal of resistor **R1** is coupled to the reference voltage **PBIAS**. One terminal of capacitor **C6** is coupled to the gate of transistor **M15**, and another terminal of capacitor **C6** is coupled to the voltage supply terminal **207**. The similar combination of capacitor **C5** and resistor **R2** is coupled to the source of transistor **M10** and transistor **MN15**, shown in FIG. **7** and described below, for tail current control through transistor **MN3**.

[0041] The impedance of a capacitor is inversely proportional to the frequency. Accordingly, at low frequency, the impedances of capacitors **C4** and **C5** are relatively high and the circuit of FIG. **5** has little effect on the control of tail current $i_{sub.ptail}$. At higher frequency, however, the impedances of capacitors **C4** and **C5** are relatively low, and the circuit of FIG. **5** helps to increase the common mode gain of amplifier **200**. The equation for tail current $i_{sub.ptail}$ is:

$$[00011] i_{ptail} = g_{ds_ptail} v_{s, \text{amp}} - g_{mp_corr} \left(\frac{g_{mc}}{g_{mx}} \right) \left(\frac{1}{1 + s/\omega_c} \right) v_{ts} \quad (\text{Eq. 15})$$

where $g_{sub.mp, \text{corr}}$ is the $g_{sub.m}$ of transistor **M15**, $g_{sub.mc}$ is the $g_{sub.m}$ of transistor **M12**, $g_{sub.mx}$ is the $g_{sub.m}$ of transistor **M11**, and

$$[00012] \omega_c = \frac{g_{ms}}{(C6 + C_{para})}$$

where $g_{sub.ms}$ is the $g_{sub.m}$ of transistor **M10**. Accordingly, per Eq. (15), at higher frequencies ($\omega_{sub.c}$ is higher), $i_{sub.ptail}$ advantageously is smaller.

[0042] FIG. **6** is a schematic diagram in which the contributions of the examples of FIGS. **3-5** are

combined as a tail current control circuit **250** that reduces the common mode gain across a wide range of frequencies. The circuitry of FIG. **6** includes the circuitry of the example tail current control circuits **250** of FIGS. **3-5** combined together in the tail current control circuit **250** shown in FIG. **6**. The source of transistor **M4** is coupled to the gate of transistor **M5**, as described above with respect to FIG. **3**, and to gate of transistor **M12**, as described above with respect to FIG. **5**. The advantages of tail current control circuit **250** include the advantages described above with respect to FIGS. **3-5**.

[0043] FIG. **7** is a schematic diagram of tail current control circuit **252** to control the tail current $i_{sub,ntail}$ through transistor **MN3**. The circuitry of tail current control circuit **252** is largely the same as that described above with respect to tail current control circuits **250** in FIGS. **3-6** but with PFETs replaced with NFETs, and vice versa. For example, PFETs **M5**, **M6**, **M7**, **M8**, and **M15** in FIG. **6** are replaced in FIG. **7** with NFETs **MN5**, **MN6**, **MN7**, **MN8**, and **MN15**. Similarly, NFET **M4** in FIG. **6** is replaced with PFET **MP4** in FIG. **7**. Tail current control circuit **252** also includes a sampling circuit **714**. Sampling circuit **714** includes a capacitor **C37** coupled to a switch **SW27**. Switch **SW27** is coupled between the gate of transistor **MP4** and the sources of transistors **MN1** and **MN2**. Sampling circuit **714** samples the voltage at the sources of transistors **MN1** and **MN2** when switch **SW27** is on, which occurs during the hold phase, as described above. The transistors **MN5-MN8** and **MP4** in FIG. **7** are connected similarly to the corresponding transistors in FIG. **6**. The gate of transistor **MN3** in FIG. **7** is coupled to the gate of transistor **MN6**. The drain and source of transistor **MN15** is coupled to the drain and source, respectively, of transistor **MN3**. The gate of transistor **MN15** is coupled to capacitor **C5** in FIG. **6** to thereby receive the correction voltage $v_{sub,gp,corr}$ for higher frequency operation, as described above.

[0044] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0045] Also, in this description, the recitation “based on” means “based at least in part on.” Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

[0046] A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

[0047] As used herein, the terms “terminal”, “node”, “interconnection”, “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

[0048] A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

[0049] While the use of particular transistors is described herein, other transistors (or equivalent devices) may be used instead with little or no change to the remaining circuitry. For example, a field effect transistor (“FET”) (such as an n-channel FET (NFET) or a p-channel FET (PFET)), a bipolar junction transistor (BJT—e.g., NPN transistor or PNP transistor), an insulated gate bipolar transistor (IGBT), and/or a junction field effect transistor (JFET) may be used in place of or in conjunction with the devices described herein. The transistors may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors or other types of device structure transistors. Furthermore, the devices may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate (GaAs).

[0050] References may be made in the claims to a transistor's control input and its current terminals. In the context of a FET, the control input is the gate, and the current terminals are the drain and source. In the context of a BJT, the control input is the base, and the current terminals are the collector and emitter.

[0051] References herein to a FET being “ON” or “enabled” means that the conduction channel of the FET is present and drain current may flow through the FET. References herein to a FET being “OFF” or “disabled” means that the conduction channel is not present so drain current does not flow through the FET. An “OFF” FET, however, may have current flowing through the transistor's body-diode.

[0052] Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

[0053] While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

[0054] Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

[0055] Modifications are possible in the described examples, and other examples are possible, within the scope of the claims.

Claims

1. An amplifier circuit, comprising: a first transistor having a terminal; a second transistor having a terminal coupled to the terminal of the first transistor; a third transistor having a first terminal

coupled to the terminals of the first and second transistors and having a control terminal; a fourth transistor having a control terminal coupled to the terminals of the first and second transistors and to the first terminal of the third transistor, the fourth transistor having a second terminal; a fifth transistor having a control terminal coupled to the second terminal of the fourth transistor and having second and third terminals; and a sixth transistor having a control terminal coupled to the control terminal of the third transistor and to the third terminal of the fifth transistor, the sixth transistor having a second terminal coupled to the second terminal of the fifth transistor.

2. The amplifier circuit of claim 1, further comprising: a first current source circuit coupled to the second terminal of the fourth transistor; and a second current source circuit coupled to the third terminal of the fifth transistor.

3. The amplifier circuit of claim 1, further comprising a capacitor coupled between the control terminal of the fourth transistor and a voltage supply terminal.

4. The amplifier circuit of claim 3, further comprising a switch coupled between the terminals of the first and second transistors and the control terminal of the fourth transistor.

5. The amplifier circuit of claim 1, wherein the first transistor is a first field effect transistor (FET), the second transistor is a second FET, and the terminals of the first and second transistors are source terminals of the first and second FETs.

6. The amplifier circuit of claim 1, further comprising a seventh transistor having a terminal coupled to the second terminal of the sixth transistor and having a control terminal; an eighth transistor having a control terminal coupled to the control terminal of the seventh transistor; and a current source circuit coupled to the eighth transistor.

7. The amplifier circuit of claim 1, further comprising: a seventh transistor having a first terminal and a control terminal, the first terminal of the seventh transistor coupled to the first terminal of the third transistor; an eighth transistor having a first terminal; a current source circuit coupled between the first terminal of the eighth transistor and a first voltage supply terminal; and a capacitor coupled between the first terminal of the eighth transistor and the control terminal of the seventh transistor.

8. The amplifier circuit of claim 7, wherein the eighth transistor has a control terminal, and the amplifier circuit further comprises: a diode-connected transistor coupled between a voltage supply terminal and the control terminal of the eighth transistor; and a ninth transistor having a first terminal coupled to the diode-connected transistor and having a control terminal coupled to the second terminal of the fourth transistor.

9. The amplifier circuit of claim 8, wherein the ninth transistor has a second terminal, and the amplifier circuit further comprises a fixed voltage circuit having an output coupled to the second terminal of the ninth transistor.

10. The amplifier circuit of claim 6, further comprising: a ninth transistor having a first terminal and a control terminal, the first terminal of the ninth transistor coupled to the first terminal of the third transistor; a tenth transistor having a first terminal; a current source circuit coupled between the first terminal of the tenth transistor and a first voltage supply terminal; and a capacitor coupled between the first terminal of the tenth transistor and the control terminal of the ninth transistor.

11. A data converter, comprising: a first stage circuit having an output; a second stage circuit having an input coupled to the output; an amplifier having an input coupled to the output of the first stage and having an output coupled to the input of the second stage, the amplifier including: a first transistor having a first terminal and having a control terminal coupled to the input of the amplifier; a second transistor having a first terminal coupled to the first terminal of the first transistor and having a control terminal coupled to the input of the amplifier; a third transistor having a first terminal coupled to the first terminals of the first and second transistors and having a control terminal; a fourth transistor having a control terminal coupled to the first terminals of the first and second transistors and to the first terminal of the third transistor, the fourth transistor having a second terminal; a fifth transistor having a control terminal coupled to the second terminal of the fourth transistor and having second and third terminals; and a sixth transistor having a

control terminal coupled to the control terminal of the third transistor and to the third terminal of the fifth transistor, the sixth transistor having a second terminal coupled to the second terminal of the fifth transistor.

12. The data converter of claim 11, further comprising: a first current source circuit coupled to the second terminal of the fourth transistor; and a second current source circuit coupled to the third terminal of the fifth transistor.

13. The data converter of claim 11, further comprising a capacitor coupled between the control terminal of the fourth transistor and a voltage supply terminal.

14. The data converter of claim 13, further comprising a switch coupled between the first terminals of the first and second transistors and the control terminal of the fourth transistor.

15. The data converter of claim 11, wherein the first transistor is a first field effect transistor (FET), the second transistor is a second FET, and the first terminals of the first and second transistors are source terminals of the first and second FETs.

16. The data converter of claim 11, further comprising a seventh transistor having a terminal coupled to the second terminal of the sixth transistor and having a control terminal; an eighth transistor having a control terminal coupled to the control terminal of the seventh transistor; and a current source circuit coupled to the eighth transistor.

17. The data converter of claim 11, further comprising: a seventh transistor having a first terminal and a control terminal, the first terminal of the seventh transistor coupled to the first terminal of the third transistor; an eighth transistor having a first terminal; a current source circuit coupled between the first terminal of the eighth transistor and a first voltage supply terminal; and a capacitor coupled between the first terminal of the eighth transistor and the control terminal of the seventh transistor.

18. A data converter, comprising: a first stage circuit having an output; a second stage circuit having an input coupled to the output; an amplifier having an input coupled to the output of the first stage circuit and having an output coupled to the input of the second stage, the amplifier including: a first transistor having a first terminal and having a control terminal coupled to the input of the amplifier; a second transistor having a first terminal coupled to the first terminal of the first transistor and having a control terminal coupled to the input of the amplifier; a third transistor having a first terminal coupled to the first terminals of the first and second transistors and having a control terminal; and a tail current control circuit having a first terminal coupled to the control terminal of the third transistor and having a second terminal coupled to the first terminals of the first and second transistors, the tail current control circuit configured to adjust a tail current through the third transistor based on a voltage at the first terminals of the first and second transistors.

19. The data converter of claim 18, wherein the tail current control circuit comprises: a fourth transistor having a control terminal coupled to the first terminals of the first and second transistors and to the first terminal of the third transistor, the fourth transistor having a second terminal; a fifth transistor having a control terminal coupled to the second terminal of the fourth transistor and having second and third terminals; and a sixth transistor having a control terminal coupled to the control terminal of the third transistor and to the third terminal of the fifth transistor, the sixth transistor having a second terminal coupled to the second terminal of the fifth transistor.

20. The data converter of claim 19, further comprising a seventh transistor having a terminal coupled to the second terminal of the sixth transistor and having a control terminal; an eighth transistor having a control terminal coupled to the control terminal of the seventh transistor; and a current source circuit coupled to the eighth transistor.

21. The data converter of claim 19, further comprising: a seventh transistor having a first terminal, a second terminal, and a control terminal; an eighth transistor having a terminal; a current source circuit coupled between the terminal of the eighth transistor and a first voltage supply terminal; and a capacitor coupled between the terminal of the eighth transistor and the control terminal of the seventh transistor.
